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EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,151	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2456	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-37 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☑ Claim(s) 34-37 is/are allowed.

7) ☑ Claim(s) 1-29 is/are rejected.

8) ☑ Claim(s) 30-33 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-37*** are pending.

2. ***Claims 1-29 are rejected.***

3. ***Claims 30-33 are objected.***

4. ***Claims 34-37 are allowed.***

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

5. ***Claims 1-20 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) hereinafter Beiderbeck.

As per claim 1, Beiderbeck discloses an apparatus, comprising:

- a parameter acquisition circuit structured to interpret a vehicle parameter value (Beiderbeck, abstract, paragraph 0035, 0083, wherein the sensor data 10 and the operating parameters 30 are captured. The operating parameters of the motor vehicle, is captured and the overtaking information is adapted; wherein sensor data and/or the operating parameters can include the vehicle parameter value); and

- a property translation circuit structured to interpret a property request value, the property request value defining, at least in part, a requested vehicle property (Beiderbeck, abstract, paragraph 0023, wherein there is provision for the overtaking information to be produced on the basis of training data, with at least one of the following properties that characterize the driver. The training data thus have at least one of the properties. As such, the property can be available for example as a speed of travel relative to the mandatory speed, as a safety distance when pulling back in after an effected overtaking maneuver or as a distance from the target vehicle when pulling out before an effected overtaking maneuver. Thus, the overtaking information including at least one of the property of the speed of travel, the safety distance and the distance from the target vehicle can be the property request values as claimed);

- ***a parameter conditioning circuit structured to generate, in response to the property request value, vehicle property data from the vehicle parameter value, the vehicle property data corresponding to the requested vehicle property*** (Beiderbeck, abstract, paragraph 0028-0029, 0031, there is preferably provision for a current operating parameter of the motor vehicle to be captured and for the operating parameter and the overtaking information to be used to recognize an intention of the driver to overtake, and for the control signal to be selected on the basis of the recognized intention to overtake. The operating parameter can be concordant with the properties of the training data, for example. As such, for example the operating parameter allows a determined distance behavior or a determined acceleration behavior to be captured that has a level of similarity with the overtaking information. To this end, the operating parameter can be compared with the overtaking information and then for example categorized as similar or dissimilar on the basis of the comparison. If the operating parameter is assessed as similar to the overtaking information, this then allows the inference that the driver intends to

overtake, for example. On the basis of the recognized intention to overtake, for example the control signal allows advice to be output to the driver indicating that the overtaking maneuver would be possible with less risk at a specific distance or else that the present overtaking opportunity should be taken because there is no overtaking opportunity at all in the next journey section}***; and***

- ***a vehicle property provisioning circuit structured to transmit the vehicle property data to an external device in response to the property request value*** (Beiderbeck, paragraph 0037, wherein there is preferably provision for the overtaking information adapted during operation of the motor vehicle to be transferred to at least one motor-vehicle-external device. The motor-vehicle-external device can be in the form of an external server or else multiple external servers, for example. The adapted overtaking information can thus be transferred to what is known as a computer cloud, that is to say to a distributed computer network, for example. The device can be available in stationary, distributed or central fashion in this case. The transfer of the adapted overtaking information to the device allows it to be sent in turn to further motor vehicles);

As pre claim 2, claim 1 is incorporated and Beiderbeck discloses wherein the apparatus further comprises a policy acquisition circuit structured to interpret a policy (Beiderbeck, paragraph 0065, wherein overtaking information 15 is produced in particular on the basis of training data that have a property characterizing the driver 13. The property can be a speed of travel 23 of the motor vehicle 1 relative to a mandatory speed 24, for example. The mandatory speed 24 is indicated for example on a road sign 25 arranged beside the road 3; wherein the policy that the speed is relative to the mandatory speed can be the policy as claimed); and the parameter conditioning circuit is further structured to generate the vehicle property data from the vehicle parameter value by translating at least a portion of the vehicle parameter value

based at least in part on the policy (Beiderbeck, paragraph 0065, wherein overtaking information 15 is produced in particular on the basis of training data that have a property characterizing the driver 13. The property can be a speed of travel 23 of the motor vehicle 1 relative to a mandatory speed 24, for example. The mandatory speed 24 is indicated for example on a road sign 25 arranged beside the road 3; wherein the policy that the speed is relative to the mandatory speed can be the policy as claimed);

As pre claim 4, claim 1 is incorporated and Beiderbeck discloses wherein the parameter acquisition circuit is further structured to receive the vehicle parameter value from the external device (Beiderbeck, abstract, paragraph 0035, 0083, wherein the sensor data 10 and the operating parameters 30 are captured. The operating parameters of the motor vehicle, is captured and the overtaking information is adapted; wherein sensor data and/or the operating parameters can include the vehicle parameter value);

As pre claim 5, claim 1 is incorporated and Beiderbeck discloses wherein the apparatus further comprises a policy acquisition circuit structured to interpret a policy that indicates the vehicle property data is to be transmitted to the external device (Beiderbeck, paragraph 0037, wherein there is preferably provision for the overtaking information adapted during operation of the motor vehicle to be transferred to at least one motor-vehicle-external device. The motor-vehicle-external device can be in the form of an external server or else multiple external servers, for example. The adapted overtaking information can thus be transferred to what is known as a computer cloud, that is to say to a distributed computer network, for example. The device can be available in stationary, distributed or central fashion in this case. The transfer of the adapted overtaking information to the device allows it to be sent in turn to further motor vehicles);

As pre claim 6, claim 5 is incorporated and Beiderbeck discloses **wherein the external device is distinct from a device that generated the property request value** (Beiderbeck, paragraph 0037, wherein there is preferably provision for the overtaking information adapted during operation of the motor vehicle to be transferred to at least one motor-vehicle-external device. The motor-vehicle-external device can be in the form of an external server or else multiple external servers, for example. The adapted overtaking information can thus be transferred to what is known as a computer cloud, that is to say to a distributed computer network, for example. The device can be available in stationary, distributed or central fashion in this case. The transfer of the adapted overtaking information to the device allows it to be sent in turn to further motor vehicles);

As pre claim 9, claim 1 is incorporated and Beiderbeck discloses **wherein the vehicle parameter value directly corresponds to the requested vehicle property** (Beiderbeck, paragraph 0031, wherein the operating parameter can be compared with the overtaking information and then for example categorized as similar or dissimilar on the basis of the comparison. If the operating parameter is assessed as similar to the overtaking information, this then allows the inference that the driver intends to overtake, for example. On the basis of the recognized intention to overtake, for example the control signal allows advice to be output to the driver indicating that the overtaking maneuver would be possible with less risk at a specific distance or else that the present overtaking opportunity should be taken because there is no overtaking opportunity at all in the next journey section)

As pre claim 10, claim 9 is incorporated and Beiderbeck discloses wherein the vehicle parameter value includes at least one of: a vehicle speed value; a prime mover speed value; a prime mover torque value; a user actuated vehicle feature value; a vehicle location value; or an image of a vehicle surrounding (Beiderbeck, paragraph 0031, 0082, The operating parameter can be concordant with the properties of the training data, for example. As such, for example the operating parameter allows a determined distance behavior or a determined acceleration behavior to be captured that has a level of similarity with the overtaking information. The driver characteristic 12 can be captured for example by the interior camera 44, the microphone 47, a seat adjustment of the seat 41, a radio receiver or a fingerprint reader. The driver characteristic 12 may alternatively be the driving style of the driver 13 himself/herself or else a characteristic actuation of an operator control element of the motor vehicle 1, in particular a turn signal lever, an accelerator pedal or a brake pedal);

As pre claim 11, claim 9 is incorporated and Beiderbeck discloses wherein the vehicle parameter value includes at least one of: a network utilization value for a network zone of a vehicle corresponding to the vehicle parameter value; a raw network message from a network zone of the vehicle; a network address for an endpoint on a network zone of the vehicle; a memory storage description of a controller of the vehicle; a value from an endpoint on a controller area network (CAN); a value from an endpoint on a local interconnect network (LIN); a value from an endpoint on an Ethernet network; or an intermediate control value (Beiderbeck, paragraph 0024, 0028, wherein the property provided for can for example also be a pulling-in angle, an acceleration profile, a utilization level of the power reserve of the motor vehicle, a distance behavior in regard to the target vehicle before the overtaking maneuver or at

the beginning of the overtaking maneuver, an acceleration behavior, a shifting behavior, a braking behavior, an approach speed behavior in regard to the target vehicle};

As pre claim 12, claim 9 is incorporated and Beiderbeck discloses wherein the requested vehicle property includes at least one of: a component temperature value; a sensor raw value; a component speed value; or an actuator feedback value {Beiderbeck, paragraph 0029, 0031, wherein there is preferably provision for the overtaking information to be produced on the basis of a current weather condition and/or a current time of day and/or a current day of the week. Preferably, the overtaking information can also be produced on the basis of a current traffic density and/or a current road category. The current weather condition can involve for example prescribing whether it is raining, whether it is snowing, what the outside temperatures are, and then again what the overtaking tendency of the driver is on the basis of the weather condition};

As pre claim 13, claim 9 is incorporated and Beiderbeck discloses wherein the requested vehicle property includes at least one of: a drivetrain component speed value; a drive shaft speed value; a drive shaft torque value; a selected gear value; a battery state of health value; a battery state of charge value; or a battery power throughput value {Beiderbeck, paragraph 0024, 0028,wherein the property provided for can for example also be a pulling-in angle, an acceleration profile, a utilization level of the power reserve of the motor vehicle, a distance behavior in regard to the target vehicle before the overtaking maneuver or at the beginning of the overtaking maneuver, an acceleration behavior, a shifting behavior, a braking behavior, an approach speed behavior in regard to the target vehicle};

Claims 14-15, 17-19, 25-29 are rejected under the same rationale as claims 1-2, 4-6 and 9-13.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. ***Claim***(s) **3, 7-8,** 16 and 20-21 is/are rejected under 35 U.S.C. 103 as being unpatentable over Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) in view of Duri et al (Pub. No.: US 2004/0267410 A1).

As pre claim 3, claim 2 is incorporated and Beiderbeck does not explicitly disclose wherein: the property request value includes an indicator of a requesting entity; the policy includes an authorization value; the apparatus further includes a requestor verification circuit structured to determine an entity authorization value based at least in part on the indicator of the requesting entity and the authorization value; and the vehicle property provisioning circuit is further structured to transmit the vehicle property data to the external device based at least in part on the entity authorization value. However, Duri discloses wherein: the property request value includes an indicator of a requesting entity; the policy includes an authorization value; the apparatus further includes a requestor verification circuit structured

to determine an entity authorization value based at least in part on the indicator of the requesting entity and the authorization value; and the vehicle property provisioning circuit is further structured to transmit the vehicle property data to the external device based at least in part on the entity authorization value (Duri, paragraph 0040-0041, 0043 wherein he PAD 210 can be tasked with requestor verification, whether the requestor is a data owner or an authorized data user. The PERM 215 can query the PAD 210 for authorization to respond to received data requests. After receiving authorization, the PERM 215 can read data from the profile data store 225.and a corresponding privacy policy in the policy data store 230. The PERM 215 then can return information complying with the consulted privacy policy to the requestor via the responder 205).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of Duri such that a requestor verification circuit is used as claimed because this would have provided a way to improve the security of the system by ensuring that the data will only sent to authorized entities.

As pre claim 7, claim 1 is incorporated and Beiderbeck does not explicitly disclose a requestor verification circuit structured to determine, in response to and based at least in part on the property request value, an entity authorization value; and a vehicle property provisioning circuit structured to transmit the vehicle property data in response to the entity authorization value. However, Duri discloses requestor verification circuit structured to determine, in response to and based at least in part on the property request value, an entity authorization value; and a vehicle property provisioning circuit structured to transmit the vehicle property data in response to the entity authorization value (Duri, paragraph 0040-

0041, 0043 wherein the PAD 210 can be tasked with requestor verification, whether the requestor is a data owner or an authorized data user. The PERM 215 can query the PAD 210 for authorization to respond to received data requests. After receiving authorization, the PERM 215 can read data from the profile data store 225.and a corresponding privacy policy in the policy data store 230. The PERM 215 then can return information complying with the consulted privacy policy to the requestor via the responder 205).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of Duri such that a requestor verification circuit is used as claimed because this would have provided a way to improve the security of the system by ensuring that the data will only sent to authorized entities.

As pre claim 8, claim 7 is incorporated and Duri further discloses a subscription circuit structured to add a requesting entity to a subscriber data list in response to the property request value; wherein the vehicle property provisioning circuit is further structured to transmit the vehicle property data to the requesting entity in response to the subscriber data list (Duri, paragraph 0042, 0045, 0059 wherein in addition to supporting request/response protocol, the PERM 215 also can support the subscription model for data. For example, according to one embodiment, each user or data owner can have a different privacy policy for each service to which that user and/or vehicle subscribes. Alternatively, a single privacy policy can be used for each user. That privacy policy can specify how that user's telematics data is to be shared and with which ASP's. In any case, data owners can formulate privacy policies by specifying whether particular items of telematics data such as diagnostic data, location information, vehicle identification numbers, and the like are to be released to selected ASP's and under what circumstances).

Claims 16 and 20-21 are rejected under the same rationale as claims 3 and 7-8.

7. ***Claim***(s) **22-24** is/are rejected under 35 U.S.C. 103 as being unpatentable over Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) in view of BARRETT et al (Pub. No.: US 2019/0394089 A1).

As pre claim 22, claim 14 is incorporated and Beiderbeck further discloses generating the vehicle parameter value by at least one of the first and the second vehicle sensors (Beiderbeck, paragraph 0017, wherein surrounding area of the motor vehicle is captured and the sensor data are provided. The surrounding area can be captured for example using a wide variety of ambient sensors of the motor vehicle, for example a camera, a radar sensor, a lidar sensor, a laser scanner, an ultrasonic sensor or a combination of these sensors).
Beiderbeck does not explicitly disclose regulating communications between a first network zone having a first vehicle sensor and a second network zone having a second vehicle sensor.
However, BARRETT discloses regulating communications between a first network zone having a first vehicle sensor and a second network zone having a second vehicle sensor (BARRETT, Fig 1-2, paragraph 0021,wherein the network node can include application software that can be executed by one or more processors. The network nodes can include ECUs (e.g., ECUs 103, 113, 123, 143, 153), domain controllers (DCs, e.g., DCs 133), sensors, actuators, or any other types of electronic devices or computers that are capable of data communications. Different network nodes can have different network interfaces that communicate over different network protocols. In some implementations, the internal network 100 can be a heterogeneous network.

For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay, and Ethernet).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of BARRETT to regulate communications between a first network zone having a first vehicle sensor and a second network zone having a second vehicle sensor as claimed because this would have provided a way to improve the security of the system by having different network zones wherein each zone has different security/privacy policies to securely and efficiently communicate data when needed.

As pre claim 23, claim 22 is incorporated and Beiderbeck discloses wherein: interpreting a vehicle parameter value comprises interpreting a plurality of vehicle parameter values (Beiderbeck, abstract, paragraph 0035, 0083, wherein the sensor data 10 and the operating parameters 30 are captured. The operating parameters of the motor vehicle, is captured and the overtaking information is adapted; wherein sensor data and/or the operating parameters can include the plurality of vehicle parameter values);

- generating vehicle property data is based, at least in part, on the plurality of vehicle parameter values (Beiderbeck, abstract, paragraph 0023, wherein there is provision for the overtaking information to be produced on the basis of training data, with at least one of the following properties that characterize the driver. The training data thus have at least one of the properties. As such, the property can be available for example as a speed of travel relative to the mandatory speed, as a safety distance when pulling back in after an effected overtaking maneuver or as a distance from the target vehicle when pulling out before an effected overtaking maneuver); and

- **a first vehicle parameter value of the plurality is from the first vehicle sensor and a second vehicle parameter value of the plurality is from the second vehicle sensor** (Beiderbeck, paragraph 0017, wherein surrounding area of the motor vehicle is captured and the sensor data are provided. The surrounding area can be captured for example using a wide variety of ambient sensors of the motor vehicle, for example a camera, a radar sensor, a lidar sensor, a laser scanner, an ultrasonic sensor or a combination of these sensors);

As pre claim 24, claim 22 is incorporated and BARRETT discloses **wherein the first network zone and the second network zone are of distinct types** (BARRETT, Fig 1-2, paragraph 0021,wherein the network nodes can include ECUs (e.g., ECUs 103, 113, 123, 143, 153), domain controllers (DCs, e.g., DCs 133), sensors, actuators, or any other types of electronic devices or computers that are capable of data communications. Different network nodes can have different network interfaces that communicate over different network protocols. In some implementations, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay, and Ethernet);

Allowable Subject Matter

8. Claims 30-33 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached on 7:30 am - 3:30 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Umar Cheema can be reached on (571) 270-3037. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2456